A Finite Black-Hole Information Ledger for Observer-Patch Holography

The OPH collaboration

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Abstract

We formulate a finite information ledger for observer-patch models using machine-checked dynamics and record channels. A continuous one-parameter star-automorphism group on a finite-dimensional private algebra acts by blockwise unitary conjugation. For a supplied projective partition, the public record channel is the exact uniform mixture of independent sign unitaries. Classical relative-entropy contraction and finite fluctuation identities supply complementary thermodynamic inputs. A declared rich-fibre adapter constructs noncommutative regional algebras with commuting disjoint blocks and unique restriction gluing. We state the support-aware entropy identity and its proposed thermality and information-relocation consequences as finite conjectures. Together these results and conjectures make the black-hole information question precise within the supplied finite model.

Finite-model scope and theorem boundary

Statements labeled *Theorem* are machine-checked in the Lean library at the cited modules. Statements labeled *Conjecture* are the explicitly unproved statements; nothing in this paper should be read as claiming them. The phrase "black hole" names the shape of the question, not a constructed physical object: the model has no continuum horizon, no gravitational collapse, no Hawking flux, and no evaporation endpoint. The bridge from the finite ledger to physical black-hole statements requires a faithful continuum realization of the finite source-causal carrier, a source-realized clock, Einstein-branch geometry and stress, and, for any observable, a prospectively fixed measurement contract under immutable custody. No observable prediction follows from this finite ledger.

1 The question in finite form

The black-hole information problem sets two commitments against each other: microscopic dynamics that preserves information, and a macroscopic record that looks exactly thermal. In the observer-patch model both commitments are formal objects, and the question becomes finite: given that ambient dynamics on a finite private algebra is unitary, and given that what any observer can hold is a public record produced by a specific channel, account exactly for the information that the record does not carry.

The model supplies sharper versions of both sides than a generic quantum system would.

First, within the supplied pointwise-continuous star-automorphism class, unitarity is not an additional assumption. The Stone-converse packet proves that every pointwise-continuous one-parameter star-automorphism group of a unital finite-dimensional private star subalgebra is blockwise unitary conjugation generated by one self-adjoint Hamiltonian per block. The theorem makes no claim about dynamics outside that class.

Second, the record channel is not a generic decoherence map. For every projective partition, the publicization pinching is exactly the uniform average of the 2^k independently signed block reflections. The channel that produces the public record is therefore an explicit uniform mixture of unitaries, with the mixing unitaries named. It is unital; entropy nondecrease for this channel is part of Conjecture 3.1.

The conjectured ledger combines these two facts with the exact thermodynamic theorems. It would identify the information absent from the public record with observer-inaccessible coherences of the ambient state through the formula in Conjecture 3.1.

2 Machine-checked inputs

Each input below is a machine-checked result in the repository, cited by module. Their joint consequence and the precise conjectures are stated in Section 3.

Theorem 2.1 (Forced unitarity; finite Stone packet). *Every continuous one-parameter group of star automorphisms of a unital, finite-dimensional private star subalgebra containing the ambient identity acts by blockwise unitary conjugation. On each Wedderburn block it is generated by a fixed self-adjoint Hamiltonian, unique up to a real scalar.*

Theorem 2.2 (Publicization is a uniform mixture of sign unitaries). *For every supplied k-block projective partition, the pinching map is the uniform average of the 2^k diagonal sign unitaries (`EventAlgebra/RecordMajorization.lean`); averaging only the two global signs does not pinch, so the independence of the signs is load-bearing.*

Theorem 2.3 (Exact data processing and second law). *Every stochastic kernel preserving a faithful stationary reference contracts Kullback–Leibler divergence to that reference (`kl_push_le`), the conditional heat bath obeys the second law (`heatBath_secondLaw`). For finite real-valued distributions and kernels, the integral fluctuation identity holds under its positivity, normalization and stationary-reference hypotheses; the pointwise Crooks identity additionally assumes detailed balance. The proofs are in `Lean/Thermodynamics/FiniteConditionalRepair.lean` and `Lean/Thermodynamics/FluctuationTheorems.lean`.*

Theorem 2.4 (Support boundary for entropy architecture). *The totalized continuous-functional-calculus logarithm sends every projection to zero, so the raw finite trace formula for relative entropy assigns the value zero to an orthogonal pair of density projections whose support inclusion fails (`EventAlgebra/SpectralEntropyBoundary.lean`). A support-aware extended-value layer is therefore mandatory, not stylistic.*

Theorem 2.5 (Conditional rich-fibre regional packet). *The locally pinned fresh-seed source run supplies split-fibre labels at four pairwise-disjoint observer windows. A declared finite adapter constructs a genuinely noncommutative block algebra with a nonunital two-by-two designated-fibre matrix corner at every observer region, elementwise commutation across distinct constructed blocks, computed four-window coverage of the declared top algebra, and unique restriction gluing with its reconstruction corollary (`QFT/RichFibreWitness.lean`, `QFT/RichFibreRegionalNet.lean`). The* *source does not produce the matrix units, state, regional restrictions, or repair, and the corner is neither a tensor factor nor a `TensorSplitReceipt`. A source-attached regional-factor theorem requires an operator algebra and a justified factor or local-channel adapter.*

Theorem 2.6 (Finite KMS identity, degenerate inheritance boundary)**.** *The finite algebraic KMS identity holds for Gibbs states of support-graded flows, and the truncation-8 nets inherit KMS structure only degenerately (`QFT/StructuralInheritance.lean`). The nondegenerate continuation on the rich-fibre net is Conjecture 3.6.*

3 Finite conjectures

Throughout, ρ is a finite density matrix on the ambient algebra, $\mathcal{E}$ the pinching of a supplied projective partition, S the support-aware von Neumann entropy, and $D(\cdot\,\|\,\cdot)$ the support-aware relative entropy. Conjectures 3.1–3.6 state the additional finite hypotheses; they do not follow from the preceding theorems.

Conjecture 3.1 (Support-aware entropy core)**.** *There are finite support-aware definitions of $S(\rho)$ and $D(\rho\,\|\,\sigma)$ obeying the basic entropy laws. For pinching they satisfy*

$$S(\mathcal{E}(\rho)) - S(\rho) = D(\rho\,\|\,\mathcal{E}(\rho)) \geq 0,$$

with equality exactly at the public fixed states, where ($\mathcal{E}(\rho) = \rho$), and the sign-unitary mixture is entropy-nondecreasing. This core is the subset of the spectral-information chain required by the ledger; the more general majorization, maximum-entropy, and H-theorem scope is not used here.

Conjecture 3.2 (Finite no-loss statement)**.** *Under the specified ambient repair dynamics the global state evolves by the forced unitary conjugation of the Stone packet. Ordinary entropy of the state is preserved; when both arguments co-evolve under the same unitary, each explicitly named jointly unitarily invariant quantity is also preserved (including support-aware relative entropy and trace distance when those quantities are defined). The statement quantifies over the specified dynamics class; no toy evolution is substituted and no arbitrary coordinate-dependent divergence is covered.*

Conjecture 3.3 (Ledger identity and relocation)**.** *For the specified record channel, the public entropy gain equals $D(\rho\,\|\,\mathcal{E}(\rho))$; the quantity is nonnegative, vanishes exactly at the public fixed states, and is carried entirely by the off-block coherences that no access cut of the specified net exposes. Information is relocated, never destroyed, and the relocation is exactly quantified.*

Conjecture 3.4 (Generalized-second-law-shaped monotone)**.** *The composition of the cap first law with Kullback–Leibler contraction yields one exact monotone combining record entropy and the capacity ledger under repair, with finite constants and a negative control exhibiting failure of the monotone when unitality of the record channel is dropped.*

Conjecture 3.5 (Complementarity receipts)**.** *There is a composition of the generic operational access-cut interface, the supplied rich-fibre data, and a justified operator/local-channel adapter that forms a two-observer ledger in which each accessible algebra determines its own record entropy, the declared block-commutation theorem supplies an algebraic locality receipt, and a glued top observable is uniquely reconstructed within the declared restriction system. No physical no-signalling or refinement-natural interpretation follows before that attachment.*

Conjecture 3.6 (Nondegenerate finite KMS receipt)**.** *The modular data of a restricted state on one genuinely noncommutative rich-fibre region satisfies the finite KMS identity at a capacity-linked inverse temperature, nondegenerately. This receipt supplies a finite KMS input but does not establish structural QFT thermality.*

Conjecture 3.7 (Finite Page-crossover inequality)**.** *For a bipartition of the specified finite net, the exact* 2^k *sign-unitary average obeys a finite inequality that locates the crossover of accessible-record information without Haar integration.*

Remark 3.8 (Conditional finite implication). Conjectures 3.1–3.6 would imply simultaneous ambient unitarity and a thermal public record, with the inaccessible information equal to the computable functional $D(\rho \,\|\, \mathcal{E}(\rho))$ and localized in observer-inaccessible coherences. The implication supplies no physical black-hole attachment. A source-selected order embedding with calibrated density (a causal-set faithful embedding), manifoldlikeness, and a source-realized clock are additional hypotheses. Any smooth Einstein reading also requires same-family tensor-curvature convergence or the independent continuum small-ball/null-balance identification.

4 Relation to standard approaches

The Page analysis, the Hayden–Preskill protocol, and island computations concern continuum or holographic systems with gravitational dynamics. The finite ledger formulates their information question using separately proved unitary dynamics and a specified record channel. Its proposed entropy and thermality composition remains conditional on Conjectures 3.1–3.6. Within the supplied automorphism class, the Stone converse gives unitary implementation; the public channel is the explicit sign-unitary mixture. A physical black-hole interpretation additionally requires the spacetime, horizon and stress attachments stated above.

5 Observables discipline

The finite ledger implies no observable black-hole signature. A proposed observable claim derived from the ledger, such as saturation-regime relaxation structure or record-capacity signatures in compact-object transients, requires a prospectively fixed clock, frame, scale, and readout contract under immutable custody before any data comparison.

A conditional instrument template is defined for the integer-k ringdown transition comb. It is not preregistered or frozen and therefore supports no observable claim. The transition rule is an imported continuation premise, not a source-derived consequence of the finite ledger. For an emission it assumes positive integer record dimensions satisfying $d_{\text{before}} = k d_{\text{after}}$, so k must divide the initial dimension. The signed black-hole entropy change is $\ln d_{\text{after}} - \ln d_{\text{before}} = -\ln k$; the positive entropy loss entering the emitted-energy formula is $\ln k$. Conditional on that rule, candidate features above the rotation line sit at $x_k = \ln k/(8\pi)$, with $x = (GM/(c^3 g(\chi)))(\omega - m\Omega_H)$ and $g(\chi) = 2\sqrt{1-\chi^2}/(1+\sqrt{1-\chi^2})$. Their offset-subtracted ratios equal $\ln k_a/\ln k_b$ under a common scale and offset for a denominator tooth $k_b \geq 2$, whose offset-subtracted value is proved nonzero. Observed frequencies must use the detector-frame remnant mass $M_{\text{det}} = (1+z)M_{\text{source}}$ in both terms; the ratio is redshift-free only with this consistent frame convention. The tooth relation uses the leading small-transition Kerr first law; finite-step background corrections are not derived by this template.

The declared linewidth-to-spacing ratio is $64\pi^2 p_0/(a g(\chi)^2 \ln k)$, which is mass-independent but spin-dependent; a constant Page coefficient is not controlled near extremal spin. The declared $(k-1)/k$ KMS factor is a within-line net-response factor, not a cross-k transition probability or prior. The algebraic invariance, divisibility and entropy-sign boundary, ladder bracketings, producer, independent byte-identical verifier, and tests are target-blind. No event likelihood has been evaluated. Published posterior samples alone are neither a strain likelihood nor a model evidence. The draft is not anchorable until a source-derived transition, common detector likelihood or sufficient likelihood product, detector-frame nuisance model, normalized integer prior or explicit finite submodel, event selection, trials, greybody normalization, no-comb reference, and convergence rule are fixed. The computed $2 \le k \le 12$ ladder is only a finite candidate submodel; excluding it would not reject the unrestricted integer family.

6 Verification surface

The proved theorems cited in Section 2 carry `#print axioms` receipts with only the standard axioms, no admissions, and no native-evaluation shortcut, in the modules named inline. The support-aware entropy statements use the general spectral-information definitions. Conjectures 3.1–3.6 are not proved by those receipts.

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